Moumen Nermine

Final-year Industrial Chemistry Engineering Student

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EDUCATION

National Institute of Applied Science and Technology (INSAT) 2022–Present

Industrial Chemistry student, specializing in Process and Environmental Engineering, with a solid training in **chemical catalysis**, electrochemical kinetics and materials science, complemented by foundations in analytical chemistry, organic chemistry and laboratory HSE. Interested in green chemistry, biomass valorization and **catalytic materials**. Seeking a 6-month final-year internship.

PROFESSIONAL EXPERIENCE

SEGULA Technologies - Internship in Electrochemical Modeling of Batteries 06/2026-08/2026

- Modeling of a **Li-ion** cell (NMC811/Graphite-SiOx) using the Single Particle Model (SPM) implemented in MATLAB.
- Optimization of the mesh and diffusion coefficient values, reducing the **RMSE from 202.13 mV to 40.5 mV.**
- Validation of the results through an **iterative approach** and a **critical analysis** against experimental data and COMSOL.

Control Union - Quality Control Internship 08/2025–09/2025

- Analysis of lubricants (viscosity and heavy metals by **ICP-OES**).
- Preparation and interpretation of analysis reports to verify compliance with ASTM D5185 and **ISO 17025** standards.

ETAP - Internship in Analytical Characterization of Hydrocarbons and Water Treatment 07/2024–08/2024

- Training in organic geochemistry techniques (**GC**, GC-MS, Rock-Eval pyrolysis) for the characterization of hydrocarbons.
- Physico-chemical analyses of water (pH, salinity, bicarbonates, sulfates, chlorides) by ion chromatography and ICP-AES.

ACADEMIC PROJECTS

Molecularly Imprinted Polymers (MIPs) for Pharmaceutical Applications

- Synthesis and characterization of a sulpiride-selective MIP by radical polymerization, with **Soxhlet extraction**, monitored by **UV-Vis spectrophotometry** and validated by controlled-release **kinetics** (240 min).
- Synthesis of a **MOF/polymer core-shell MIP** for targeted delivery of doxorubicin, with an **85% encapsulation efficiency**.
- **Modeling of adsorption isotherms** (bi-Langmuir, Freundlich) to relate material properties, binding sites and performance.

Techno-Environmental Analysis of Ethylene Production: Steam Cracking vs Bioethanol Dehydration

- Simulation of both routes in Aspen Plus V14, including bioethanol dehydration over an **HZSM-5 acid catalyst**.
- Cradle-to-gate LCA (OpenLCA, ISO 14040/14044) on 5 impact categories; identification of the origin of the **biomass** and indirect land-use change as the main uncertainties.

Literature Review: Synthesis and Characterization of Superconducting Materials

- Analysis of the fundamental properties of superconductors and study of fabrication routes: solid-state ceramic synthesis, crystal growth **(flux, Bridgman, Czochralski)**, and thin-film deposition **(ALD)**.
- Study of characterization methods: **RAMAN, FTIR, XRD, SEM and TEM**.

Prediction of Liquid–Liquid Equilibria for Ternary Extraction Systems

- Formulation of the LLE thermodynamic system and resolution via Rachford-Rice and **arc-length continuation**.
- Implementation of the workflow in Python, MATLAB and Mathematica (spline/fsolve, SciPy/mpltern, Interpolation/FindRoot).

TECHNICAL SKILLS

- **Laboratory Techniques:** Electrochemistry (Cyclic voltammetry, polarography), Chromatography (HPLC, GC, TLC), Spectrometry (MS, ICP-OES), Spectroscopy (NMR, FTIR, UV-Vis), Diffraction (XRD), Microscopy (SEM, TEM).
- **Simulation:** Aspen Plus and HYSYS, COMSOL Multiphysics • **Programming:** $MATLAB, Mathematica, Python$
- **Data & Office Tools:** Microsoft Office Suite, OpenLCA • **Standards:**$ISO9001, ISO14001, ISO45001, ISO17025, ISO14040$

EXTRACURRICULAR ACTIVITIES

IEEE INSAT — Organizing Chair of TRSYP 3.0; Social Activities Chair – MENASYP; Chair of EMBS.
THEATRO INSAT — Member of the dance team.

LANGUAGES AND CERTIFICATES

Languages
English (Advanced), French (Bilingual), Arabic (Native).

Certificates
Aspen HYSYS – MATLAB